

FlowState: Benchmarking Deployment Capacity and Alpha Decay in Indian Equities under Observable Information

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Abstract

We present **FlowState**, a frozen benchmark that measures two properties of an equity trading strategy from daily bars alone: how much capital it can deploy, and how fast its edge decays. Both are reported for a rules-based Indian liquidity universe of 184 symbols over 2020–2024.

The benchmark’s central design decision is a negative one. Deployment capacity is normally defined as the AUM at which a strategy’s own trading erodes its returns to zero, which requires a calibrated market-impact function. We measure whether daily bars can supply one and report that **they cannot, and that the obstruction is identification rather than precision**: the sign of the estimated transient-impact coefficient depends on the weighting scheme applied across symbols, with precision weighting and equal weighting disagreeing at 3 of 4 horizons. FlowState therefore reports a *constraint* figure — the AUM at which a strategy can no longer trade within a stated participation limit — which is arithmetic on observed turnover and assumes nothing.

Three findings follow, all null or negative. Deployment capacity for standard price-and-volume factors is **Rs. 0.81–7.91 crore**, set in every case by the least liquid position the strategy holds. **No factor’s edge decays measurably** within a quarter; the largest $|t|$ at the shortest horizon is 1.79. And deployable size **does not collapse** when foreign participation reverses — the effect the benchmark was designed to detect — with outflow-to-inflow ratios between 0.94 and 1.24. That last null rests on a derivatives-based flow proxy and cannot distinguish a stable capacity from an uninformative proxy; we say so in the results rather than the discussion.

Applying the frozen benchmark to 156 machine-generated strategies before publishing it exposed two defects that 5 standard factor strategies had not, and the corrections moved every capacity figure *down* by between a factor of four and a factor of twenty-six. We report that as a result about benchmark validation, not as an implementation note.

1 What this benchmark refuses to measure

A portfolio manager’s first question about a strategy is how much money it can take. The standard answer is a *capacity curve*: net return as a function of deployed AUM, falling as the strategy’s own trading pushes prices against it, with capacity defined as the AUM at which the edge reaches zero. Constructing one requires knowing how much a given order moves the price — a market impact function.

We could not obtain one, and Section 3.4 is the measurement rather than the assertion. The obstruction is not that daily bars are noisy. It is that a daily bar aggregates every participant’s trading and cannot isolate one order, and — more decisively — that temporary impact decays over minutes to hours, so the finest resolution a daily bar permits is already coarser than the phenomenon being estimated.

Rather than fit an impact model to data that cannot identify one, FlowState answers a smaller question honestly:

Constraint-based deployment capacity. The largest AUM at which a strategy can place every one of its trades without exceeding 1.0% of a session’s traded value. This is arithmetic on observed turnover. It is what a real desk’s pre-trade risk system enforces, and it is a strictly smaller claim than impact-erosion capacity — which is why it can be made at all.

Every capacity figure in this paper is a constraint figure. None is an impact-erosion figure. The two answer different questions and only one of them is supported by daily data.

Contributions

1. A frozen, reproducible benchmark reporting deployment capacity and alpha decay for Indian equities, with every published figure generated from artifacts rather than transcribed.
2. A measurement, with a detectability bound and a positive control, showing that daily OHLCV cannot identify a transient-impact coefficient — and that the failure is non-identification rather than insufficient power.
3. Deployment capacities for standard factors on an Indian liquidity universe, which we believe are not openly published.
4. A null result on flow-conditional capacity, reported with the limitation of its proxy stated alongside it.
5. A methodological result: validating a benchmark on a narrow strategy family failed to expose defects that a broader machine-generated population exposed immediately.

2 The benchmark

Universe and window. A rules-based liquidity universe — at each quarterly rebalance, the top names by trailing median traded value, computed only from sessions strictly prior. Nothing consults a published index membership list, so stocks that later delist simply stop qualifying rather than vanishing from history. Over 2020–2024 this yields 184 symbols and 122,464 investable symbol-days. A holdout year is reserved and was never opened by any part of this work.

Factors. Five signals built from price and volume alone: twelve-month momentum skipping the most recent month, one-month reversal, low realised volatility, Amihud illiquidity, and a traded-value size proxy. Books are equal-weighted quintile long-short portfolios formed every session.

Value and quality are absent, and their absence is a scope limitation rather than an oversight. Both require company fundamentals that this repository does not hold and could not acquire under a zero data budget. Any comparison against published factor results must be read knowing that the two most fundamentals-dependent families are missing. The fifth factor is named `liquidity_size_proxy` rather than `size` for the same reason: it is built from traded value, not market capitalisation, which needs a share count we do not have.

Capacity. For a book rebalancing from weights w to w' , the traded fraction in name i is $|\Delta w_i|$. At deployed AUM A the rupee trade is $A|\Delta w_i|$, and the participation constraint permits at most κ of that session’s traded value, so

$$A_{\max}(t) = \min_{i: \Delta w_i \neq 0} \frac{\kappa \cdot V_i(t)}{|\Delta w_i(t)|}, \quad A_{\max} = \min_t A_{\max}(t),$$

where $V_i(t)$ is a *trailing* average traded value — using the session’s own volume would size the order with the volume the order helped create. A strategy’s first session enters at full weight from zero, so building the opening position is priced like any other trade.

The outer minimum over sessions is deliberate and was not always there; Section 3.5 describes what taking a median instead concealed.

3 Results

3.1 Deployment capacity

Factor	Capacity	Entry	Median session	Median÷binding	One name
<code>momentum_12_1</code>	6.94	28.61	31.4	4.5	27.4
<code>reversal_1m</code>	0.81	9.89	21.2	26.2	7.7
<code>low_volatility</code>	6.78	8.31	32.9	4.8	40.1
<code>illiquidity</code>	7.22	10.75	32.0	4.4	24.1
<code>liquidity_size_proxy</code>	7.91	16.02	33.9	4.3	18.8

Table 1: Deployment capacity in Rs. crore at a 1.0% participation limit. Capacity is the binding session; entry is the session that builds the position; the median session and its ratio to the binding figure are shown to indicate what a typical-session summary would have reported. *One name* is the share of sessions on which the tightest position is more than twice as tight as the next.

Deployment capacity is **Rs. 0.81–7.91 crore**, roughly one to ten million US dollars. **The least liquid position decides it.**

Why the least liquid position decides it. These are equal-weighted books, so every name in a leg carries the same weight and must be traded in the same rupee size. A strategy holding a name that turns over a tenth of what another turns over trades in the same size as it; the participation constraint binds on the thinner one first, and the whole book’s capacity is set there. Capacity is therefore a function of the *least* liquid name a strategy holds, not of the average liquidity of its book — which is why a strategy trading a hundred large caps can still have a capacity of a few crore. This is not an artefact of our definition. It is the constraint a desk actually faces, and the standard remedy — capitalisation weighting, or a liquidity screen on entry — changes the portfolio and therefore changes the returns the capacity is a capacity *for*.

Dropping the single tightest position each session raises capacity by well under an order of magnitude, which is what establishes that these figures describe the strategy rather than one holding.

3.2 Alpha decay

0 of 5 factors have a measurable half-life. Every curve is flat to gently rising out to a quarter.

We report this as a null rather than as a finding about India, because the precondition fails: the largest $|t|$ at the shortest horizon is 1.79, so the edges whose decay is being measured are not themselves established. The largest $|t|$ anywhere in the table, at any horizon, is

Factor	Return per session held (bps)				<i>t</i> -statistic			
	<i>h</i> =1	<i>h</i> =5	<i>h</i> =21	<i>h</i> =63	<i>h</i> =1	<i>h</i> =5	<i>h</i> =21	<i>h</i> =63
<code>illiquidity</code>	+4.86	+4.96	+5.62	+6.82	+1.79	+1.75	+2.07	+2.40
<code>momentum_12_1</code>	+3.96	+3.83	+5.15	+5.94	+1.12	+1.08	+1.50	+1.95
<code>liquidity_size_proxy</code>	+2.57	+2.44	+3.02	+3.88	+1.22	+1.14	+1.49	+2.12
<code>low_volatility</code>	−3.02	−3.79	−4.92	−5.33	−0.73	−0.93	−1.28	−1.39
<code>reversal_1m</code>	−6.22	−6.59	−3.40	−1.74	−1.79	−2.04	−1.05	−0.50

Table 2: Long-short return per session held, gross of costs, across 1,233 formation dates. *t*-statistics are corrected for the overlap that daily formation induces by dividing the sample by the number of independent holding periods.

`sDecayMaxAbsTstat`. A decay half-life is only a meaningful quantity for a signal that had an edge, and reporting one for a factor that never made money would be a category error dressed as a measurement.

Two factors carry the wrong sign for their names. `reversal_1m` loses money, so this window shows one-month *continuation* rather than reversal; `low_volatility` loses, so high-beta names outperformed — consistent with a window containing the COVID crash and the recovery that followed it.

A consequence for the research question. FlowState was specified to ask whether alpha decays faster in India than in developed markets and whether retail participation explains it. The first clause is answered — no decay was observed — and the second is therefore retired rather than deferred: there is nothing for retail participation to explain. Delivery-percentage data that would have supported the retail analysis is freely available and was deliberately not collected, on the principle that the availability of data is not a reason to ask a question.

3.3 Flow-conditional capacity

Outflow-to-inflow capacity ratios lie between 0.94 and 1.24 across 5 factors, median 0.96, on 76–234 sessions per cell. **The hypothesis is not supported.**

FlowState’s novel contribution was to ask whether deployable size collapses when foreign participation reverses — a plausible order-of-magnitude effect in a market with large and volatile foreign flows. It does not move materially, and some ratios point the wrong way.

This null rests on a proxy, and we cannot separate two explanations. No free source serves historical daily FII/DII *cash-market* net flow; NSE’s endpoint returns only the current session regardless of the date requested. The series used is FII *equity-derivatives* activity, which differs in three ways that are not small: it is denominated in contracts rather than rupees, over a window in which regulatory lot sizes changed; it describes derivatives rather than cash; and FII derivatives activity is substantially hedging against cash positions, so its sign need not track the direction of cash-market demand. Whether capacity is genuinely stable across flow states, or the proxy is simply uninformative about cash liquidity, is not separable with this data. We state this here rather than in a limitations section because it governs how the null may be read.

3.4 Why daily OHLCV cannot price transient impact

This is a principal finding of the benchmark, not a caveat on it.

Horizon (sessions)	Precision-weighted β	Equal-weighted β	MDE pooled	MDE one symbol
1	−0.0180	−0.0114	0.0135	0.182
5	−0.0395	+0.0205	0.0267	0.368
10	−0.0588	+0.0370	0.0350	0.497
21	−0.0996	+0.0867	0.0477	0.699

Table 3: Reversal regressions of the following h -session return on today’s, restricted to each symbol’s heaviest-volume decile, $n = 177$ symbols. MDE is the minimum effect distinguishable from zero at 80% power.

A price move caused by an order is *transient*: it is the cost of demanding liquidity and it is repaid once the demand stops. A move caused by information is permanent. A negative reversal coefficient is therefore evidence of impact, and the sign of that coefficient is the quantity a daily-bar impact estimate depends on.

The sign depends on the weighting scheme. Precision weighting and equal weighting are both standard, neither is wrong, and they **disagree on sign at 3 of 4 horizons**. A quantity whose sign is set by an analyst’s choice among defensible estimators is not identified by the data.

Three further measurements agree, and one control shows the failure is in the model rather than the data:

- **Resolution.** Temporary impact decays over minutes to hours; the finest horizon a daily bar permits is one session. The pooled coefficient also grows monotonically more negative with horizon rather than decaying, which is the signature of slow mean reversion rather than of a liquidity concession being repaid.
- **Exponent.** Fitting $|r|$ against relative traded value gives a median exponent of 0.689 against the square-root law’s 0.5, spanning 0.521–0.859 across 179 symbols and explaining a median 0.126 of variance. Across a sample split the per-symbol exponent correlates 0.374 with itself ($n = 166$).
- **Extrapolation.** A capacity study asks about 1.0% participation. Observed relative traded value has a median of 0.821, 1.91 orders of magnitude away, and only 2 of 208,042 symbol-days fall inside the region of interest.
- **The control.** Amihud illiquidity, computed on the same data, correlates 0.836 across the same split ($n = 166$, $p = 1.7 \times 10^{-44}$). The data supports a stable liquidity *measure* while refusing to support an impact *model parameter*.

Two corrections to our own earlier reporting. An earlier version of this analysis reported that *no* reversal was detected, citing the proportion of symbols with a negative coefficient as evidence. That proportion was noise: a single symbol’s daily test can detect only an effect of 0.368 or larger at the five-session horizon, an order of magnitude above the pooled coefficient of -0.0395 that the same data supports. We had reported an underpowered statistic as a measurement. Computing the detectability bound then showed that the pooled coefficient *is* significantly negative — contradicting our own earlier claim. The identification argument above replaced it and is stronger, but the sequence is worth recording: the null we first reported was not the null that survived.

3.5 The validation gap

FlowState was validated on 5 standard factor strategies spanning $9.8\times$ in capacity, then applied to 156 machine-generated strategies spanning $30.9\times$ — $3.2\times$ wider. **Two defects lived in that gap, and the corpus found both immediately.**

Before freezing the benchmark we ran it over a corpus of strategies generated by a local language model and audited by two earlier benchmarks in the same programme. The intent was to produce a reference corpus in which every strategy carries an audit verdict, a fragility score and a deployment capacity; 125 strategies do. What it produced first was two bugs.

Defect one: floating-point residue counted as trading. A strategy holding a hundred names at equal weight and never rebalancing produces weight changes of order 10^{-18} from recomputing an unchanged number. Dividing a participation limit by that residue reported its capacity as 10^{23} crore. The remedy is a tolerance, 1.0×10^{-9} , separating a trade from arithmetic noise.

Defect two: the summary statistic discarded the binding constraint. Capacity was reported as the *median* session. For a near-static book that is close to meaningless: almost every session has near-zero rebalancing turnover, so the one tightly constrained session that builds the position was drowned by a thousand that barely trade. The remedy is the minimum across sessions, which is the definition given in Section 2. Using the median instead would overstate the typical corpus strategy by $10.6\times$, and entry is the binding session for only 32 of 156 strategies (20.5%) — so the median was concealing more than the entry constraint alone.

Neither defect was reachable from the validation set. All five factor strategies rebalance substantially every session, so none of them ever produced a near-zero traded fraction, and for all five the median and the minimum differ by a factor small enough to look like a modelling choice rather than a bug.

The correction moved every published number down. Capacity figures fell by between a factor of four and a factor of twenty-six. We record the direction because it bears on whether this counts as tuning: adapting a benchmark to flatter its subject would not systematically reduce the results it reports.

The general form: **a benchmark’s soundness cannot be established on the population that motivated it.** Defects concentrate in the tails of the population a benchmark is applied to, and a validation set assembled from familiar, well-behaved strategies has no tails.

4 Threats to validity

- **T1 — The flow proxy.** The flow-conditional null rests on derivatives activity, not cash-market flow, for the reasons given above. This is the threat we would attack first.
- **T2 — Capacity is an extreme-value statistic.** The binding figure is a minimum over roughly twelve hundred sessions and is therefore sensitive to a single session. We report the fifth percentile and the drop-the-tightest-name variant alongside it, but the headline is a minimum and should be read as one.
- **T3 — The factor zoo is incomplete.** Value and quality are absent for want of fundamentals, which weakens any comparison against published factor results.
- **T4 — Decay is measured on edges that are not established.** With no factor significant at the shortest horizon, the decay curves describe the decay of quantities that may be zero.
- **T5 — Returns are gross of costs.** The decay curves do not net transaction costs, which in Indian equities are large enough to reverse the sign of a small edge.
- **T6 — One market, one window.** 2020–2024 in Indian equities contains one pandemic crash and one recovery. The negative results may be properties of that window.
- **T7 — The corrections were made after seeing the corpus.** Both defects in Section 3.5 were found by the data the benchmark was about to be applied to, and repaired before freezing. We argue this repairs objective omissions rather than tuning, and note the corrections all moved results downward, but a reader is entitled to weigh that themselves.

5 Conclusion

FlowState reports what daily bars can support and declines what they cannot. Deployment capacity for standard price-and-volume factors on an Indian liquidity universe is Rs. 0.81–7.91 crore, set by the least liquid position each strategy holds. No factor’s edge decays measurably within a quarter, and none is established at the shortest horizon. Deployable size does not collapse when foreign participation reverses, though that null is conditional on a proxy we would rather not have needed.

The result we expect to outlast the others is the one about instruments rather than about markets. Daily OHLCV cannot identify a transient-impact coefficient — not for want of precision, but because two equally defensible estimators disagree about its sign. A great many published backtests price market impact from daily data; this is a measurement of what that assumption is worth.

FlowState is intended as benchmark infrastructure rather than a completed scientific result. It is frozen, versioned, and reproducible from raw data by a documented command sequence, so that future strategy generators and future capacity methods can be evaluated against the same protocol and progress can be measured against a common standard.

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